Vemula Siddhartha

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.52
Aaditya Junior College	Class XII (TSBIE)	2023	97.60%
Iris World School	Class X (ICSE)	2021	90.83%

SCHOLASTIC ACHIEVEMENTS

- **Academic Excellence** Award for the highest GPA, IIT Hyderabad.
- **Teaching Assistant** for the courses **Probability and Random Processes, Matrix Theory** and **Deep Learning**; assisted students with problem-solving and concept clarification and graded quizzes and assignments.
- **A+** grade awarded in **7** courses (top 2% of the class, rarely given).

EXPERIENCE

- **Research Intern** | IISc Bengaluru
- Fine-tuned **feed-forward 3D Gaussian Splatting (3DGS)** models with a modified training objective, achieving **~50% reduction in Gaussian primitives** while maintaining comparable **PSNR, SSIM, and LPIPS** to the original models.
- Developed and evaluated **geometry-based Gaussian pruning** methods to reduce redundant Gaussians in feed-forward NVS; benchmarked **state-of-the-art depth estimation and reconstruction models**.
- Built an evaluation pipeline for **novel-view reconstruction**, including image preprocessing, pose alignment, Gaussian rendering with **gsplat**, and quantitative evaluation using **PSNR, SSIM, and LPIPS**.

PROJECTS

- **Inventory Optimization using Online Convex Optimization:**
- Implemented **Smoothed Online Gradient Descent (SOGD)** for online inventory optimization with non-stationary demand and convex Newsvendor loss; studied **dynamic regret** and path-length-dependent bounds.
- Replicated **MaxCOSD** and evaluated regret across single-product, perishable, and 100-product inventory settings under different learning rates, with experiments on synthetic demand sequences up to 50,000 timesteps.

- **Adaptive Noise Cancellation using RLS:**
- Developed an **RLS-based adaptive noise canceller** in MATLAB for non-stationary noise with **full and partial suppression**.
- Integrated a **notch filter for tonal noise**; achieved **~40 dB SNR gain** in full suppression and **~20 dB** in partial suppression.

- **5-Stage Pipelined RISC-V Processor:**
- Designed a **64-bit 5-stage RISC-V processor** with pipelined IF/ID/EX/MEM/WB stages, implementing arithmetic, branching, jumps, memory operations.
- Implemented **data forwarding, load-use hazard detection, pipeline stalls/flushes, and store-data forwarding** for resolving data and control hazards; deployed and validated the processor on a **PYNQ-Z2 FPGA**.

- **Two-Way OFDM Communication System:**
- Developed a **two-way OFDM transceiver** on **ADALM-PLUTO SDR** using MATLAB, implementing **BPSK/QPSK modulation, pilot-aided channel estimation, synchronization, and frequency-domain equalization**.
- Implemented **per-subcarrier adaptive modulation** based on estimated SNR, with **FFT/IFFT processing and cyclic-prefix insertion/removal** for transmission over varying channel conditions.

- **State Space Models & Kalman Filtering:**
- Studied **S4/S6 and Mamba**, focusing on **selective state updates, HiPPO initialization, and hardware-aware selective scan** for efficient long-context sequence modeling.
- Investigated **Kalman filtering/smoothing and AI-aided state estimation**, including **RTS smoothing, learned**

Kalman gains, and DNN-KF hybrid architectures; analyzing tradeoffs in interpretability, uncertainty, and adaptability across model-based and data-driven approaches.

SKILLS

- **Programming Languages:** C, C++, Python, MATLAB.
- **Libraries:** NumPy, SciPy, PyTorch, Pandas, scikit-learn.
- **Tools:** Git, Bash, LaTeX, SolidEdge, Vivado, GNU Radio, Simulink.

RELEVANT COURSES

- **Math:** Probability and Random Variables, Stochastic Processes, Matrix Theory, Convex Optimization, Numerical Methods.
- **CS:** Computer Architecture, Data Structures and Algorithms, Operating Systems, Theory of Computation, Discrete Math.
- **AI:** Machine Learning, Deep Learning, Reinforcement Learning, Speech Systems, Game Theory, AI in Finance.
- **EE:** Information Theory, Wireless Communication, Digital Signal Processing, Analog Electronics, Control Systems.

EXTRACURRICULARS

- **Position of Responsibility:** UG Mentor, Sunshine- Campus Counselling Cell.
- Represented IIT Hyderabad in **Inter-IIT Chess** Tournament; **State-Level Cricket** Player.
- **Carnatic Violinist & Vocalist; Rubik's Cube: India Top 200; Zetamac High Score: 107.**